

**KIGILYAH, Russian Federation**WMO: **216360**Lat: **73.336N**Lon: **139.869E**Elev: **79**StdP: **14.65**Time Zone: **10.00 (E10)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |      |       | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|------|-------|-----------------------|------|-----|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%   |       |                       |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS   | MCDB  | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l)  | (m)   | (n)                   | (o)  | (p) |
| 2             | -38.6      | -35.8 | -41.4                         | 0.5 | -38.2 | -38.7 | 0.6 | -35.6 | 33.3                  | -15.8 | 30.7 | -12.0 | 5.1                   | 110  | N/A |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.9                    | 53.2            | 48.9 | 49.4 | 46.4 | 46.6 | 44.3 | 49.1                | 52.5 | 46.8 | 49.3 | 44.5 | 46.2 | 9.8                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 46.7                            | 47.3 | 50.3 | 44.9 | 44.1 | 48.1 | 42.9 | 40.8 | 46.3 | 19.8          | 52.9 | 18.6 | 49.4 | 17.4 | 46.3 | 57.9              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 30.1          | 27.1 | 23.8 | DB                         | -40.4 | 56.8               | 2.6 | 3.4                                                | -42.3 | 59.2       | -43.8 | 61.2       | -45.3 | 63.1       | -47.2 | 65.6 | (4) |
| (5)               |      |      | WB                         | -40.4 | 53.2               | 2.6 | 3.2                                                | -42.3 | 55.5       | -43.8 | 57.4       | -45.3 | 59.2       | -47.2 | 61.5 | (5) |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct  | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)  | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 9.9    | -21.0 | -22.0 | -14.3 | 2.9   | 20.1  | 34.6  | 37.7  | 40.7  | 34.9  | 20.6 | -1.4  | -16.1 |
|                                                                                  | DBStd       | 24.68  | 8.25  | 9.31  | 10.73 | 8.69  | 7.96  | 5.01  | 4.32  | 4.15  | 4.66  | 8.68 | 10.00 | 8.78  |
|                                                                                  | HDD50       | 14641  | 2200  | 2016  | 1993  | 1414  | 926   | 464   | 382   | 290   | 453   | 911  | 1542  | 2050  |
|                                                                                  | HDD65       | 20112  | 2665  | 2436  | 2458  | 1864  | 1391  | 913   | 845   | 754   | 903   | 1376 | 1992  | 2515  |
|                                                                                  | CDD50       | 5      | 0     | 0     | 0     | 0     | 0     | 1     | 2     | 2     | 0     | 0    | 0     | 0     |
|                                                                                  | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0     | 0     |
|                                                                                  | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0     | 0     |
|                                                                                  | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0     | 0     |
| Wind                                                                             | WSAvg       | 11.5   | 9.2   | 10.2  | 10.8  | 12.0  | 13.6  | 12.2  | 12.0  | 12.2  | 11.9  | 13.5 | 10.5  | 10.1  |
| Precipitation                                                                    | PrecAvg     | 4.3    | 0.2   | 0.2   | 0.1   | 0.2   | 0.2   | 0.6   | 0.9   | 0.9   | 0.7   | 0.4  | 0.1   | 0.1   |
|                                                                                  | PrecMax     | 7.3    | 0.5   | 0.6   | 0.4   | 0.4   | 0.8   | 1.2   | 2.5   | 1.7   | 1.7   | 0.8  | 0.3   | 0.4   |
|                                                                                  | PrecMin     | 2.0    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.2   | 0.1  | 0.0   | 0.0   |
|                                                                                  | PrecStd     | 1.7    | 0.1   | 0.2   | 0.1   | 0.2   | 0.2   | 0.3   | 0.7   | 0.5   | 0.4   | 0.2  | 0.1   | 0.1   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 5.3   | 7.3   | 22.4  | 25.5  | 36.8  | 56.8  | 58.0  | 56.3  | 49.1  | 35.5 | 25.4  | 7.1   |
|                                                                                  |             | MCWB   | 5.2   | 6.9   | 21.5  | 24.5  | 34.9  | 50.0  | 52.8  | 52.0  | 46.4  | 34.6 | 24.4  | 6.7   |
|                                                                                  | 2%          | DB     | 0.3   | 2.0   | 17.0  | 22.0  | 34.9  | 51.1  | 53.6  | 51.3  | 45.7  | 33.9 | 22.2  | 3.4   |
|                                                                                  |             | MCWB   | -0.1  | 1.7   | 16.1  | 21.2  | 33.2  | 46.5  | 49.3  | 48.3  | 43.3  | 32.6 | 21.4  | 3.0   |
|                                                                                  | 5%          | DB     | -3.7  | -3.8  | 9.7   | 18.6  | 33.3  | 45.0  | 49.5  | 48.8  | 42.8  | 32.4 | 17.6  | -0.5  |
|                                                                                  |             | MCWB   | -4.0  | -4.0  | 9.5   | 17.7  | 32.0  | 41.4  | 46.3  | 46.4  | 41.1  | 31.2 | 17.0  | -0.9  |
|                                                                                  | 10%         | DB     | -7.8  | -8.2  | 0.2   | 15.8  | 31.9  | 41.4  | 44.9  | 46.8  | 40.9  | 30.6 | 13.0  | -3.6  |
|                                                                                  |             | MCWB   | -8.0  | -8.5  | -0.2  | 15.0  | 31.0  | 38.6  | 42.2  | 44.9  | 39.3  | 29.2 | 12.5  | -4.0  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 4.9   | 7.1   | 21.5  | 24.6  | 35.0  | 50.3  | 53.4  | 52.8  | 46.6  | 34.6 | 24.4  | 6.7   |
|                                                                                  |             | MCDB   | 4.9   | 7.3   | 22.1  | 25.1  | 36.4  | 55.9  | 56.7  | 55.9  | 48.8  | 34.9 | 25.0  | 6.8   |
|                                                                                  | 2%          | WB     | 0.1   | 1.9   | 16.1  | 21.3  | 33.3  | 46.2  | 49.6  | 48.5  | 43.1  | 32.8 | 21.4  | 3.1   |
|                                                                                  |             | MCDB   | 0.4   | 2.1   | 16.8  | 22.0  | 34.4  | 50.8  | 53.4  | 50.7  | 45.0  | 33.5 | 22.5  | 3.3   |
|                                                                                  | 5%          | WB     | -4.0  | -4.0  | 9.3   | 17.8  | 32.2  | 41.7  | 46.6  | 46.8  | 41.3  | 31.2 | 17.0  | -0.9  |
|                                                                                  |             | MCDB   | -3.6  | -3.9  | 9.6   | 18.5  | 33.1  | 44.6  | 49.9  | 48.6  | 42.6  | 32.1 | 17.7  | -0.6  |
|                                                                                  | 10%         | WB     | -8.0  | -8.4  | -0.1  | 15.2  | 30.9  | 38.6  | 42.5  | 45.1  | 39.6  | 29.5 | 12.5  | -4.0  |
|                                                                                  |             | MCDB   | -7.8  | -8.2  | 0.3   | 15.8  | 31.9  | 41.2  | 44.6  | 46.3  | 40.7  | 30.6 | 12.9  | -3.6  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.0   | 9.1   | 10.1  | 12.0  | 8.0   | 6.3   | 7.2   | 5.9   | 4.2   | 5.7  | 8.9   | 8.6   |
|                                                                                  |             | MCDBR  | 13.5  | 14.1  | 16.6  | 15.5  | 7.6   | 12.2  | 14.9  | 8.7   | 5.9   | 3.8  | 12.1  | 12.5  |
|                                                                                  |             | MCWBR  | 13.4  | 13.9  | 16.6  | 15.2  | 7.1   | 8.9   | 11.7  | 6.5   | 4.9   | 4.1  | 11.8  | 12.6  |
|                                                                                  | 5% WB       | MCDBR  | 12.8  | 14.0  | 16.6  | 15.8  | 7.7   | 12.3  | 14.9  | 8.0   | 5.6   | 3.7  | 12.1  | 12.3  |
|                                                                                  |             | MCWBR  | 12.7  | 13.9  | 16.6  | 15.6  | 7.2   | 9.3   | 12.3  | 6.5   | 5.0   | 4.1  | 11.8  | 12.4  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | N/A    | 0.173 | 0.270 | 0.303 | 0.335 | 0.327 | 0.363 | 0.353 | 0.298 | 0.357 | N/A  | N/A   |       |
|                                                                                  | taud        | N/A    | 1.888 | 1.982 | 1.993 | 1.983 | 2.188 | 2.269 | 2.407 | 2.429 | 2.727 | N/A  | N/A   |       |
|                                                                                  | Ebn at Noon | 0      | 123   | 204   | 247   | 256   | 266   | 250   | 236   | 211   | 102   | 0    | 0     |       |
|                                                                                  | Edh at Noon | 0      | 8     | 24    | 36    | 43    | 36    | 32    | 25    | 16    | 7     | 0    | 0     |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0      | 49    | 481   | 1288  | 1933  | 2043  | 1639  | 893   | 397   | 90    | 0    | 0     |       |
|                                                                                  | RadStd      | 0      | 6     | 18    | 48    | 75    | 124   | 113   | 112   | 39    | 13    | 0    | 0     |       |

**Historical Trends**

|                             |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)                         | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)   |       |       |
| (46) Station Only           | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (2 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page